HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · OIL & GAS · PLANT HSE

REFINERY WORKFORCE · TEN HAZARDS · TWO LANGUAGES

Plant Safety Video Series

Ten bite-size animated safety videos, bilingual and built with GenAI as lightweight HTML. One rule per video, playing on a gate kiosk, a toolbox-talk tablet or a phone, even offline.

~~6 wk~~ → **1** day to produce each video

10 videos
Animated safety videos
▲ core plant hazards covered

~30 sec
Length of each video
▲ one rule, one minute or less

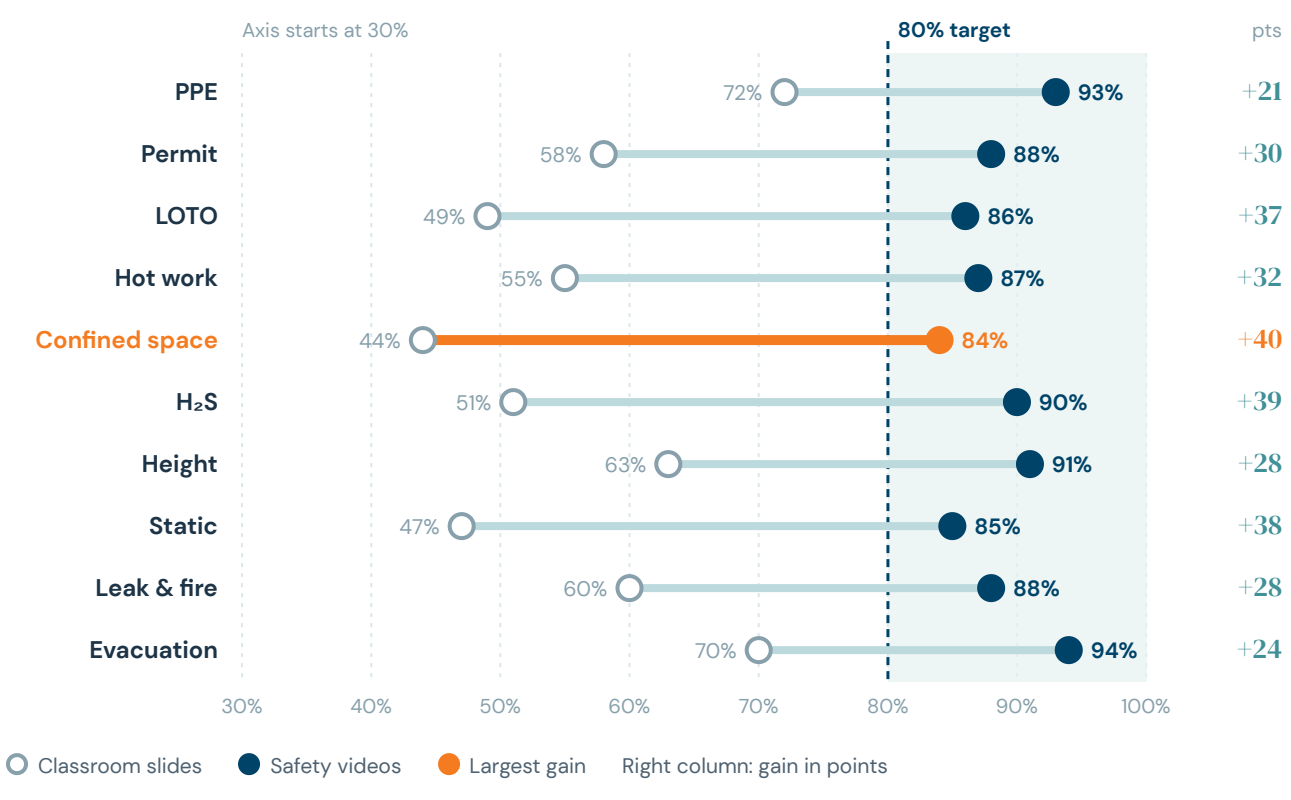
1 day
Production time per video
▼ vs ~6 weeks with an agency

2 languages
Captions & narration
▲ English and Hindi, more to follow

Rings: one segment per topic; 30 seconds against the one-minute ceiling; one day against ~6 weeks (inner ring); one segment per language. Figures indicative; validated metrics available on request.

01 Remembered, not just watched

Recall after induction · correct answers in a 3-question check, % by topic · indicative



TARGET LINE
0 → 10
Topics at or above the 80% recall target: none with classroom slides, all ten with the videos.

BIGGEST JUMP
Confined space: 44% → 84%
The densest procedure had the weakest recall, and gained the most: +40 points.

PATTERN
Densest SOPs gained most
LOTO (+37), Static (+38) and H_2S (+39) follow the same curve; simpler topics like PPE (+21) started higher.

02 Ten videos, one rule each

The series as delivered · every video ends on the same three-point "Remember" card

Title — Topic and the one rule
Scene — action (×3)
Scene — optional 4th
Remember — Three things to keep

01 PPE
Dress for the plant
The right PPE, every time.
1. All six PPE items, every time
2. Gas monitor on in process areas
3. Inspect it; replace if damaged

02 PERMIT
No permit, no work
Every non-routine job starts with a permit.
1. No valid permit, no work
2. Permit on display at the job
3. Changes? Stop and revalidate

03 LOTO
Lock it. Tag it. Try it.
Isolation before every intervention.
1. Identify every energy source
2. Your own lock and named tag
3. Try it: verify zero energy

04 HOT WORK
Hot work
Control every spark.
1. Hot work permit + gas test
2. Clear, cover and screen
3. Fire watch during and after

05 CONFINED SPACE
Confined space
Never just pop in.
1. Isolate, blind and ventilate
2. Test top, middle and bottom
3. Attendant stays; no BA, no rescue entry

06 H_2S
H_2S awareness
Don't trust your nose.
1. Never rely on smell
2. Alarm: stop, make safe, leave
3. Upwind or crosswind; no BA, no rescue

07 HEIGHT
Working at height
100% tied off, 100% of the time.
1. Green tag only
2. Harness + double lanyard, one hook always on
3. Tether tools; barricade below

08 STATIC
Earth before you load
Control static and ignition sources.
1. Spark arrestor; phones out
2. Earth and confirm green
3. Slow start; stay with the vehicle

09 LEAK & FIRE
Leak or fire
Act in the right order.
1. Alarm first, then control room
2. Isolate and contain only if safe
3. Small fire: PASS. Growing fire: leave

10 EVACUATION
When the siren sounds
Know exactly what to do.
1. Stop and make safe
2. Walk upwind to your assembly point
3. Headcount; wait for the all-clear

Five-beat structure, about 30 seconds; one caption per scene at a Class 8 reading level, in English and Hindi. Content illustrative and aligned to common refinery practice; final scripts to be validated by the plant HSE team against site SOPs before use.

03 Four pain points, four fixes

Before → after

BEFORE — Long, dense SOPs
Permit, LOTO and confined-space procedures ran to many pages, skimmed at induction and forgotten by shift two.
ONE RULE PER VIDEO
AFTER — ~30-second videos
Each SOP reduced to one core rule and three things to remember.

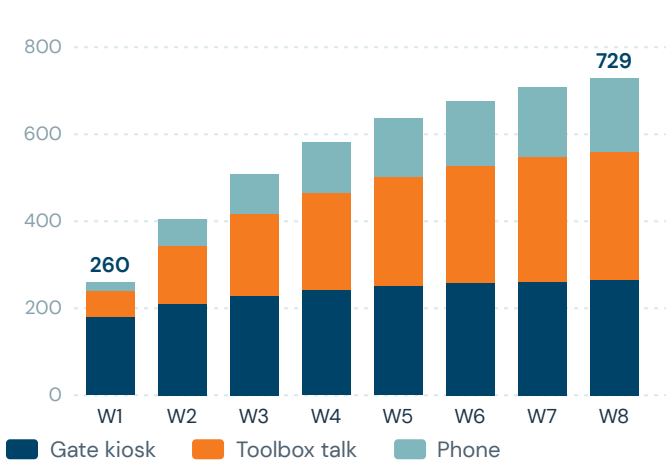
BEFORE — Video was slow and costly
Weeks of shooting and editing per film; updating one step meant re-shooting.
ANIMATED IN CODE
AFTER — A day per video
Edit a scene or caption and re-publish, the same week the procedure changes.

BEFORE — Many languages on site
A mixed workforce needed the same message spoken as well as written; material was English text only.
GENAI TRANSLATION
AFTER — English and Hindi
Captions and optional narration, switchable live; more languages to follow.

BEFORE — Low connectivity zones
Heavy files streamed poorly at gates and in process units, where training is needed most.
SINGLE HTML FILE
AFTER — Plays offline
One light file on kiosk, tablet or phone, plus a daily safety minute at toolbox talks.

04 Where it's watched

Plays per week by channel, first 8 weeks · indicative



05 Code beats camera

Production time per video, and what changes after

Agency film
~6 weeks of shooting and editing

Generated in code
about a day per video

Bars to scale: 1 day against ~42 days

CHANGE A STEP	
~~Re-shoot~~	**Edit a scene or caption**
FILE	
~~Large, streams poorly~~	**A few kilobytes, plays offline**
LANGUAGES	
~~English text only~~	**English + Hindi, voice too**

06 How HMC ran it

1 week	**Discover** — Ten high-consequence topics shortlisted from SOPs, the permit system and near-miss learnings.
1 week	**Design** — Storyboards and a bilingual caption sheet, five beats per video.
2 weeks	**Build** — Claude animated each storyboard as SVG; HSE reviewed every frame.
1 month	**Adopt** — Gate kiosks, toolbox talks and supervisors' phones; daily rota.
∞ ongoing	**Scale** — New topics and regional languages; template and prompt pack.

What transfers to any plant safety team

01 — One rule per video
Thirty seconds on a single behaviour beats thirty minutes on everything. People remember the picture, not the paragraph.

02 — Code beats camera for safety content
Animation in code is cheap to change. When a procedure changes, the video changes the same week.

03 — Say it in their language
Captions and voice in the languages spoken on site reach the contractor who needs the message most.

Built with **Claude** · **SVG + JavaScript** · **Web Speech** · **Single HTML file** | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

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